

Transition readiness

2007 stationary reference — block0506, September 28 verified export
September 28, 2026 | Preparation and validation; no policy effect claimed

Decision

The dated household and population code can be reused, but the previous transition launcher cannot be run unchanged. It selects an older calibration and an approximate credit rule. The new isolated bridge authenticates all 260 saved parameter fields and imports the frozen September 28 source. Preferences and measurements remain fixed.

Your two definitions are recorded: **fixed physical housing stock, with prices and rents clearing**; and **removal of artificial borrowing/down-payment limits while preserving repayment and lifetime solvency**. Individual housing and tenure choices remain free.

Numerical preparation

Check	Status	Evidence
Exact reference control	PASS	113 exact arrays; 14 fit rows, 31 parameters, 17 identical plots. Imported control; no new solve.
Two-date no shock	Timed out	180-second cap reached before the first dated audit. No operator acceptance.
One-date diagnostic	PASS	2 calls; 111.9 s; housing 1.76e-09; PAYGO 3.17e-13.
Six dates / fixed stock	Not run	Longer demographic and fixed-stock operator gates remain open.

The source project is read-only. Earlier failures exposed an outdated serializer and a diagnostic path pointing into the reference. The bridge now checks all fields itself and redirects only diagnostic output. No shocked equilibrium or natural-credit implementation is certified.

Meaning of the old estimated transition

The short historical fit estimated four successive unexpected preference changes: after each surprise households expected its new level to last forever. A separate announced-path experiment revealed all four changes at once. These are different expectations assumptions.

Surprise	Fertility window	Old preference	Target	Short fit
2007	2008-2011	0.129	1.975	1.973
2011	2012-2015	0.117	1.861	1.861
2015	2016-2019	0.106	1.755	1.755
2019	2020-2023	0.092	1.646	1.646

The subsequent long-horizon refit accepted **zero fitted shocks**. One 104-date market/fiscal root passed but missed fertility and remained 1.742% away in population and 9.446% in distribution distance from its endpoint. Old absolute preference levels cannot silently transfer to the new utility specification.

Closures that must remain explicit

Estimated and normalized inputs are frozen at their saved values. An endogenous outcome is solved after a shock; an outstanding closure is not silently replaced by a convenient default.

Object / classification	Maintained treatment
Preferences / estimated or calibrated, then fixed	All saved preferences, including child-benefit scale. No fertility renormalization after a credit shock.
Earnings / externally estimated	Approved B15 Markov process, 15 income states and saved age profile. Reuse the completed measurement audit.
Population / normalized initial state	Exact saved pre-choice distribution, mass one. Carry level-valued population forward; no reset or age reweighting.
Adult entry / author-fixed normalization	Half of a birth vintage enters after 16 years, half after 20; adjusted births map to households at 1/2.1. Keep raw and adjusted queues.
Entry wealth / empirical normalization	Saved conditional wealth/income distribution; retain negative entry assets and inherited rank coupling. No zero-wealth substitute.
Survival and geography / fixed	Saved age survival; one pooled nationally calibrated market. Closed diagnostic path, with no new migration, retention or old quota defaults.
PAYGO / empirical baseline, fixed tax	Hold payroll tax at 8.028%; solve dated pension benefits from actual worker and retiree masses. Baseline benefit 0.918 is a starting value.
Property tax / externally fixed	Saved annual rate 1.060%, period rate 4.239%; zero household rebate. Do not import old equal rebates.
Estate settlement / outstanding	Retain provisional net-estate funding of actual next entrants and residual sink. Shortfalls and negative estates fail. Counterparty and physical settlement remain unresolved.
Housing / author-defined experiments	Credit experiment retains the absolute elastic supply curve. Fixed-stock arm holds actual reference supply (5.848), not the intercept (6.294). Replacement of depreciation is implicit.
Terminal population / endogenous	Solve renewal, PAYGO and housing jointly. A unit-mass stationary distribution alone is not a demographic equilibrium.

Natural solvency is a separate implementation gate

Construct feasible states backward at each decision node, preserving the existing information and fertility/housing timing. Every reachable future income/family outcome must permit repayment. At each possible death, post-saving liquid wealth plus net housing liquidation value must be nonnegative. The same condition applies at terminal age. With positive death risk and no default or life insurance, it can itself exclude unsecured borrowing; positive future earnings do not guarantee repayment after an early death.

The old option infers feasibility from a value cutoff and the first feasible grid node. It conflicts with the saved incumbent-owner debt rule. Replace the artificial purchaser and incumbent limits explicitly, then verify worst-income/death boundaries and an expanded/refined debt grid. Financed share equal to one is insufficient.

Architecture, gates and finite budgets

Endpoint and path are different problems

At fixed preferences, solve the price and pension for demographic renewal and balanced PAYGO. With B adjusted births, E entrant households and d housing demand per normalized household, renewal requires $B/(2.1 E) = 1$. Housing then determines the population level $N = H^S(q)/d$; under fixed stock, replace $H^S(q)$ by the reference stock. Verify native one-step distribution and both entry queues. Do not restore fertility mechanically if a root is absent.

Calculation	Interpretation
Impact / prescribed continuation	Household response from the exact inherited population. No market-clearing claim.
Heuristic short path	A stated guessed price/pension sequence, with population carried forward. Not an equilibrium price path.
Cleared finite path	Households, housing and pensions clear at every included date. Endpoint and horizon adequacy still require verification.
Genuine transition	Finite clearing plus terminal distribution/queue approach and stable early responses under horizon extension.

A pure 10% supply-intercept increase has an algebraic endpoint candidate: unchanged prices, pension and per-household policies, with population and all aggregate flows 10% higher. Renewal, PAYGO and the provisional estate ledger scale together. Native scaling checks remain required; this does not establish uniqueness, stability or a transition.

Bounded sequence

Stage	Maximum / stop condition
Current preparation	45 active minutes, excluding queue time. Serial Torch checks, one CPU/24 GiB. After a 180-second two-date timeout, one instrumented date is capped at six minutes and two calls. No policy run.
Natural-solvency verification (proposed)	12 constructed boundary cases plus two fixed-price household solves, 20 minutes. Stop at support/repayment disagreement.
One credit endpoint (proposed)	16 stationary evaluations including repeat/one-step checks; 45 minutes. No preference search. Stop without a valid renewal bracket or at budget.
Six-date cleared diagnostic (proposed)	First smoke six dates (about 11 min; 15-min cap). Then at most eight root mappings plus one replay: 108 calls, about 94 min; proposed 2-hour cap. Count Jacobian work separately.
Historical fixed stock / long horizon	Not launched. Requires an agreed shock contract and a separate finite plan. The other chat owns the +10% supply comparison.

Use measured replay/mapping time before approving the next budget. The independent fixed-price control took about 66 seconds. The six-date mapping crosses both maturation lags; the initial entry mismatch is only 4.889e-8 households, but subsequent drift must be reported rather than repaired by normalization.

Acceptance and next decision

Preserve source/checkpoint identities, all preference fields, exact initial mass, occupied household budgets and debt feasibility, probabilities, value monotonicity, estate funding, and zero inherited-state projection. Retain housing tolerance 2e-4, PAYGO 1e-6, mass accounting 2e-8 and backward/forward reproduction 1e-10. Keep the 17 standard plots and add dated price, rent, quantities, population and boundary-distance panels.

For the historical fixed-stock comparison, the author has been asked whether to leave historical shocks pending or use old proportional preference declines only as an illustrative diagnostic. No old shocks, new estimates, preferences or fiscal closures have been adopted here.

Traceable sources: transition_readiness.md; historical_user_evidence.jsonl; historical_source_identity.json; the frozen reference manifest; September 13 original-queue and September 16 recovery receipts. The exact output paths and commands are indexed in the packet README.

Complete reference target fit

Reference loss 19.581. These are unchanged reference observations, not counterfactual results. All 14 rows are retained, including the normalization and three zero-weight validation rows.

Moment	Target	Model	Gap	Weight	Loss	Role
Completed fertility	2.1	2.1	-1.66e-06	-	-	normalization
Childlessness	0.198	0.201	0.003	3.55e+04	0.275	scored
Exactly one child	0.214	0.209	-0.004	2.70e+04	0.483	scored
Mean first-birth age	25.976	25.933	-0.043	139.828	0.264	scored
First birth at 30+	0.249	0.224	-0.026	0	0	validation
Wealth / earnings	6.927	6.326	-0.6	7.595	2.737	scored
Bequests / wealth	0.007	0.007	-2.34e-04	5.17e+06	0.282	scored
Old-age dispersion	3.516	3.069	-0.447	0	0	validation
Mean rooms	5.729	5.848	0.119	128.021	1.798	scored
Ownership ages 30-55	0.676	0.655	-0.021	2339.362	1.076	scored
First-birth rooms response	1.465	1.622	0.157	137.565	3.395	scored
Family rooms gap	0.385	0.353	-0.032	0	0	validation
Recent-parent ownership gap	0.128	0.12	-0.008	2.71e+04	1.758	scored
Children at age 25	0.81	0.535	-0.274	100	7.513	scored

The full-precision target table is preserved in the selected export. Loss comparisons require this exact target/weight system. Completed fertility was a calibration normalization; credit-policy evaluation will display its gap instead of imposing it.

Identity and interpretation

Checkpoint: b15ba92d...7309d. Source manifest: 07d84336...496d. Identification contract: 68323aad...5abf. Full hashes and all 260 serialized fields are in fixed_reference_manifest.json. Two exact repetitions and the standard plot identities were authenticated before this preparation.

Existing limitations remain: high-wealth ownership and age-30 housing profiles, buyer-conditional policy plots, held-out earnings covariance miss, and provisional estate/entry interpretation. Reproduction verifies implementation identity; it does not establish global calibration optimality or resolve these economic issues.

Estimated and normalized quantities

All are frozen for the requested credit experiment. Search bounds describe the inherited calibration; no search takes place in this preparation. A near-bound flag means within 1% of the inherited search interval.

Parameter	Estimate	Bounds / restriction	Near bound
H0	6.294	[0.2, 80]	No
beta annual	0.963	[0.94, 0.99]	No
chi	1.094	[0.1, 5]	No
first birth fixed cost	0.621	[0, 8]	No
kappa fert	0.176	[0.02, 50]	Yes
kappa fert continuation	0.332	[0.02, 50]	Yes
theta0	0.125	[0, 8]	No
delta alpha jump	0.135	[0, 0.25]	No
child benefit curvature	0.061	[0, 0.8]	No
tenure choice kappa	0.012	[0.001, 0.1]	No
psi child	0.136	Replacement normalization; now frozen	-

The two fertility taste scales are near their lower search bounds. This is an inherited calibration caveat, not permission to drop targets, change weights or renormalize preferences after a shock.

Diagnostics

The 17 standard PNGs remain unchanged in the selected-export standard_diagnostics folder. The imported exact control reproduces every PNG hash. This readiness memo does not replace that full diagnostic packet; new solved cases must produce it again, with supplemental transition panels clearly labeled.

All remaining reference parameters

Continuation of the complete 31-row parameter table. Values, restrictions and meanings come from the saved export; constructor defaults are not used.

Parameter	Value	Inherited restriction / meaning
child benefit CRRA coefficient	0.127	derived from normalized one-child benefit
theta1	0.008	fixed external restriction
sigma	2	fixed
alpha cons	0.733	fixed CEX childless expenditure share
delta alpha	0	fixed zero later-child loading
h P	0	no housing floor
utility reference rent	0.11	fixed substantive utility normalization
q annual	0.02	author-retained 2% annual real rate
financed share	0.8	inherited credit contract
housing supply elasticity	0.63	fixed provisional external mapping
payroll tax	0.08	derived from adopted pension ratio
pension period	0.918	balanced PAYGO
annual depreciation	0.014	adopted
period depreciation	0.055	compounded
annual property tax	0.011	adopted
period property tax	0.042	linear period convention
selling cost	0.06	retained
rental cap	6	retained provisional
wealth grid nodes	160	retained exact grid
income states	15	retained B15

In a counterfactual, a retained financed-share number is metadata if its operative artificial constraint is replaced. Payroll tax stays fixed but the pension is solved from dated PAYGO. Housing supply changes only in the explicitly specified fixed-stock arm.